Geophysical granular flow benchmarks over frictional and bumpy surfaces

Retief Lubbe*, Hongyang Cheng

Soil Micro Mechanics (SMM), Faculty of Engineering Technology, University of Twente

UNIVERSITY OF TWENTE.

Lorentz center

INTRODUCTION & MOTIVATION

Collapse of the Brumadinho tailings dam
2019

The dam failure released around *12 million* m^3 of tailings.
Mud flow traveled for about *50 km* and reached the river, traveling 250 km south
270 people died - Vale had to pay *7bln dollars* damages

INTRODUCTION & MOTIVATION

Problem:
Geophysical granular flows (landslides, debris flows) involve extreme deformations, transitioning from static → flowing → static

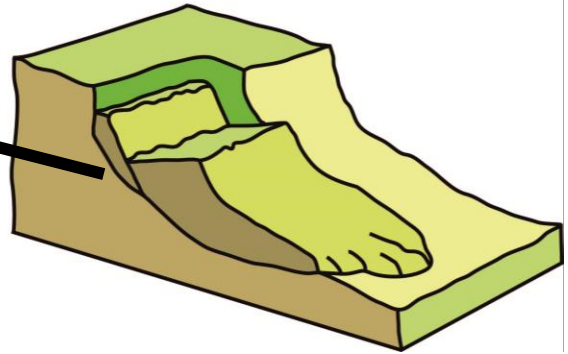
Challenge in SciML:
- High dimensionality in state space
- Changing topology
- Heterogeneity in boundary conditions (friction and roughness).

Objective:
Provide a **multi-stage benchmark** that tests **ML models'** ability to **generalize** from on-set to failure.

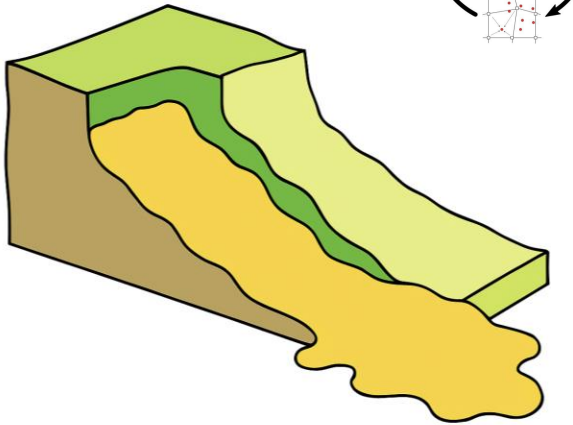
BENCHMARK STAGES

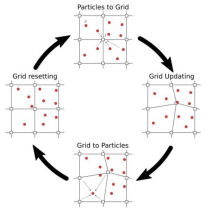
BENCHMARK STAGES

Onset of a landslide



Failure





(Stage 0)
Local stress element test
(e.g., triaxial compression)

stress-volume-history response under a load

(Stage 1)
Boundary value problem test
(e.g., Material Point Method)

Conservation laws and state evolution of the slope

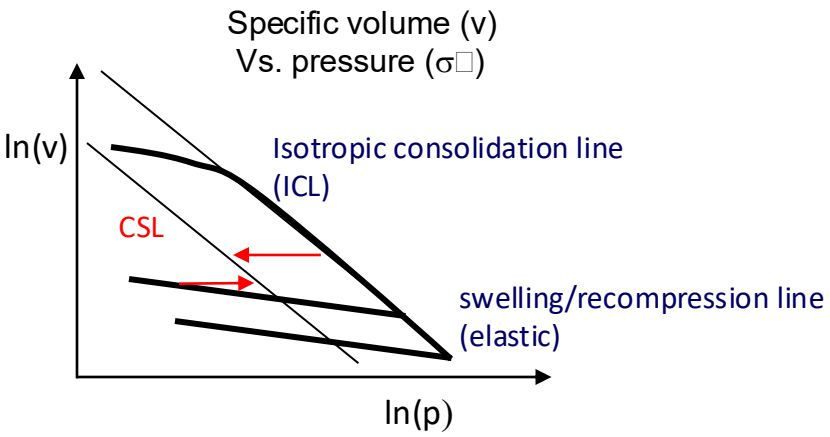
STAGE 0

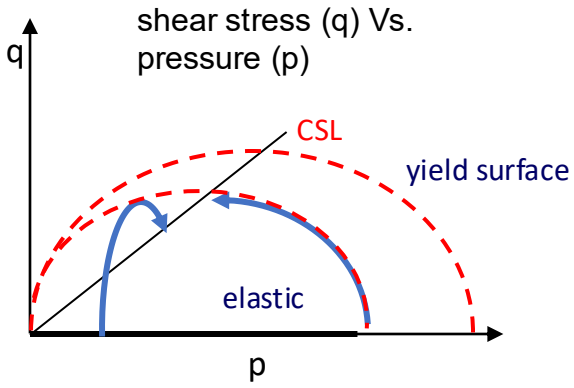
ELEMENT TEST

(ON-SET OF FAILURE)

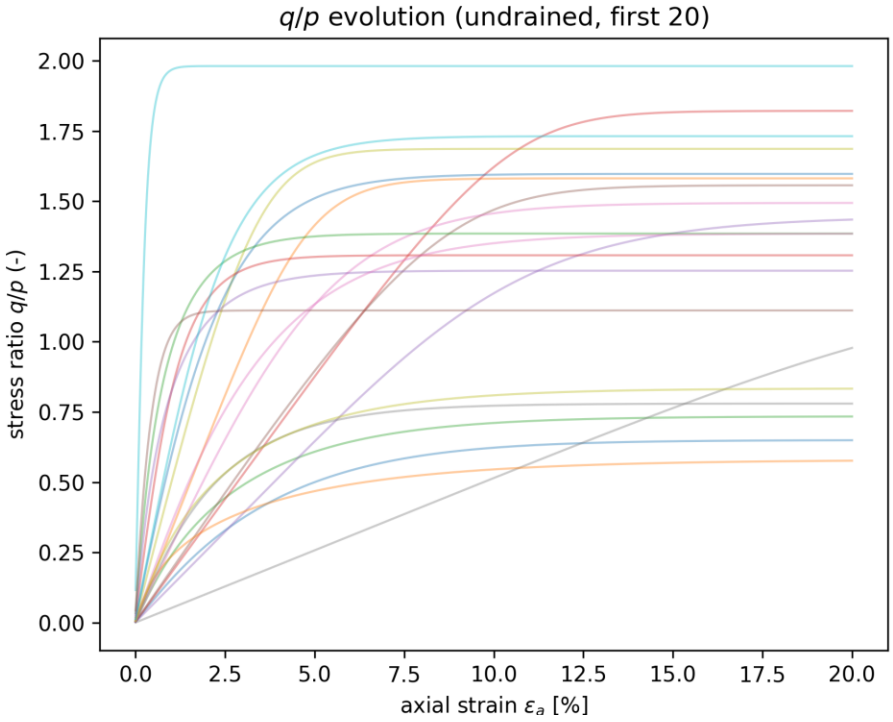
MODIFIED CAM CLAY

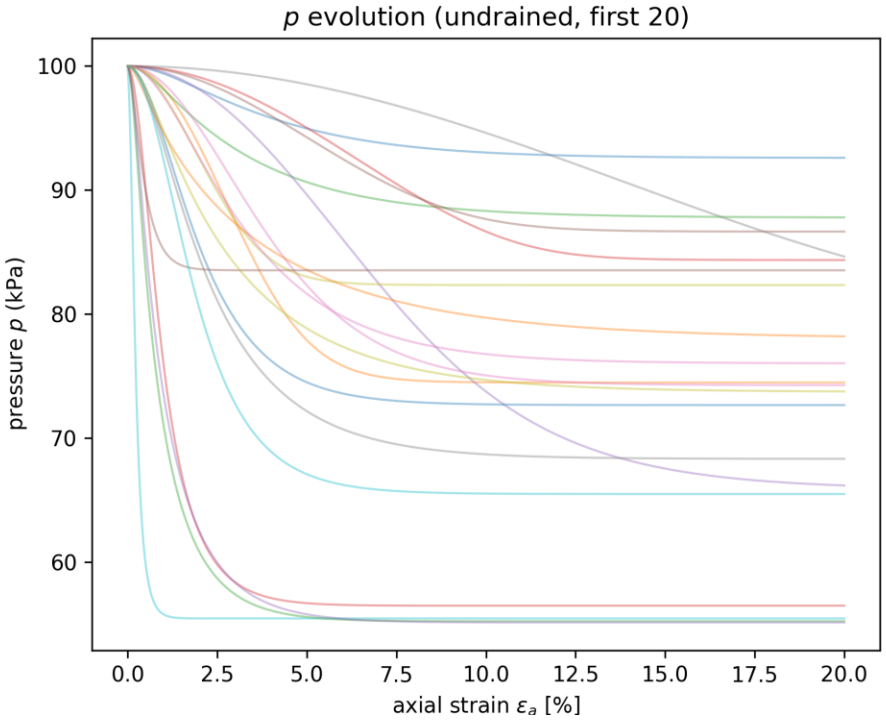
Elasto-plastic constitutive law
Soils fail under shear & stress history has a role



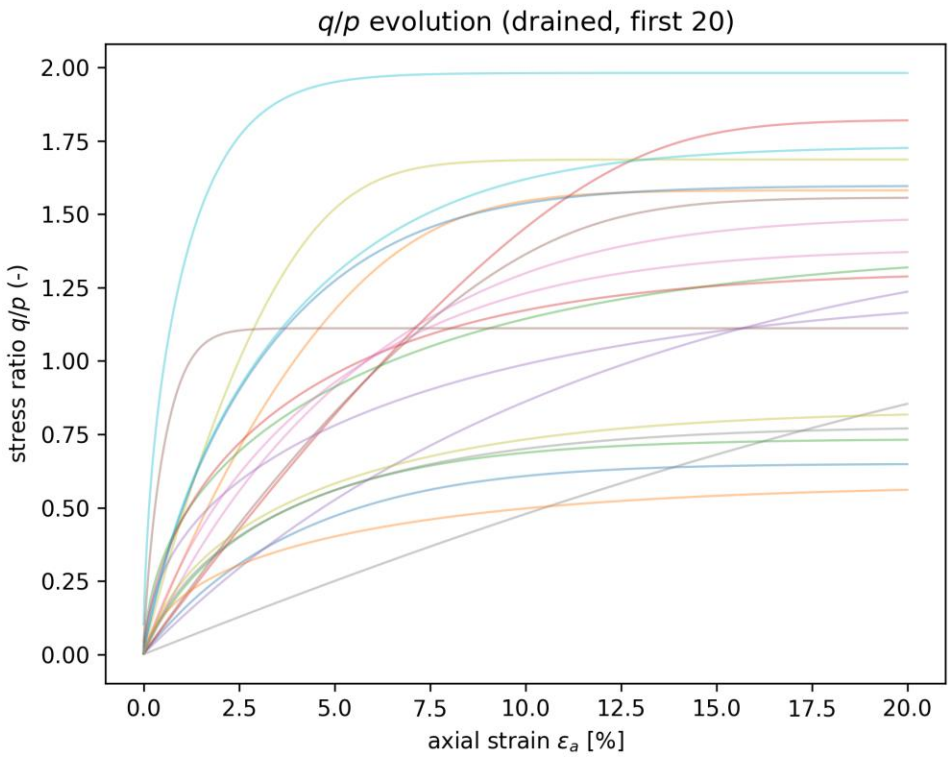


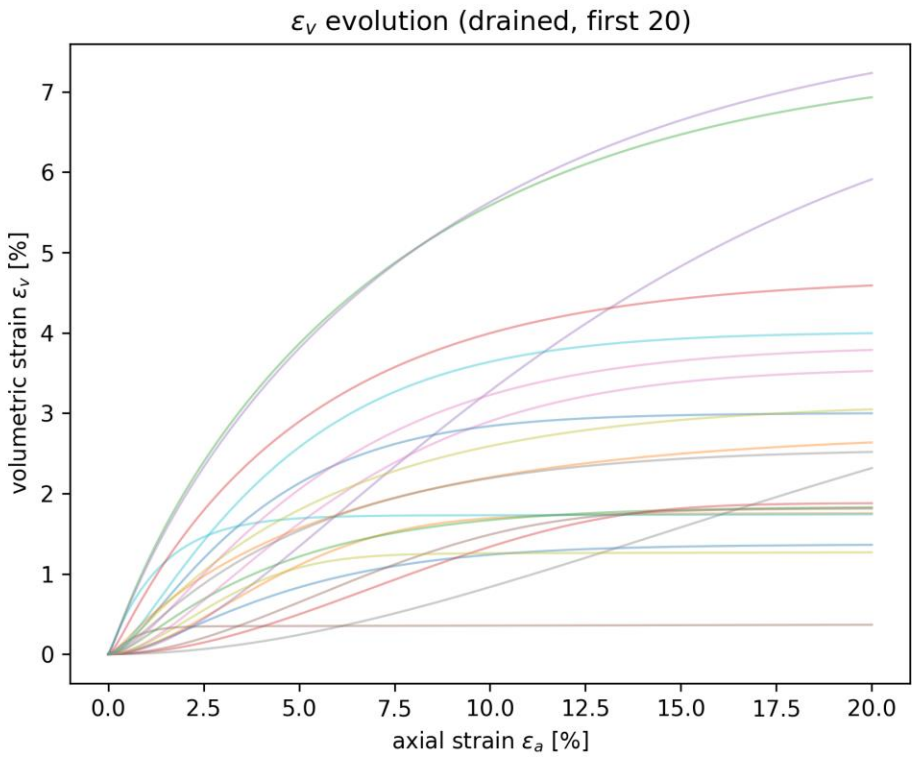
UNDRAINED TRIAXIAL COMPRESSION



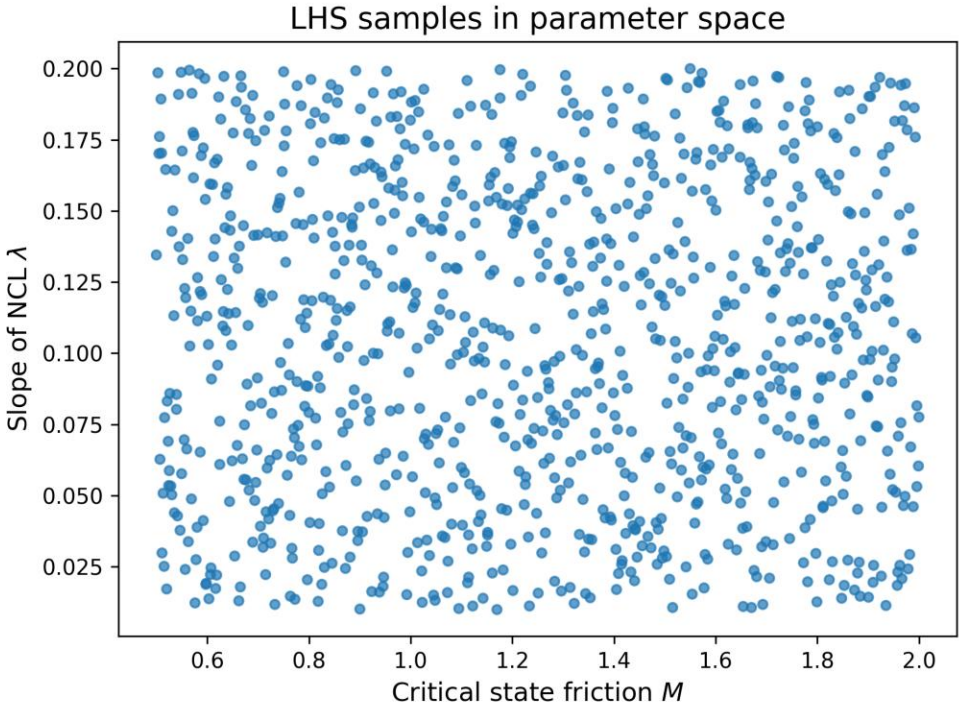


DRAINED TRIAXIAL COMPRESSION





MODEL PARAMETERS



Critical state friction *M: 0.5 to 2.0*

Slope of normal consolidation line *λ: 0.01 to 0.2*

Slope of swelling line *κ (%λ) : 0.01 to 0.2*

Specific volume of ICL *N: 0.4 to 4.0*

Poisson's ratio ν: 0.2 to 0.3

Data stored as HD5

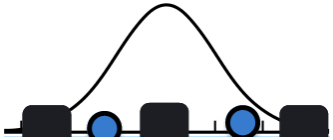
N=1000 Samples

STAGE 1

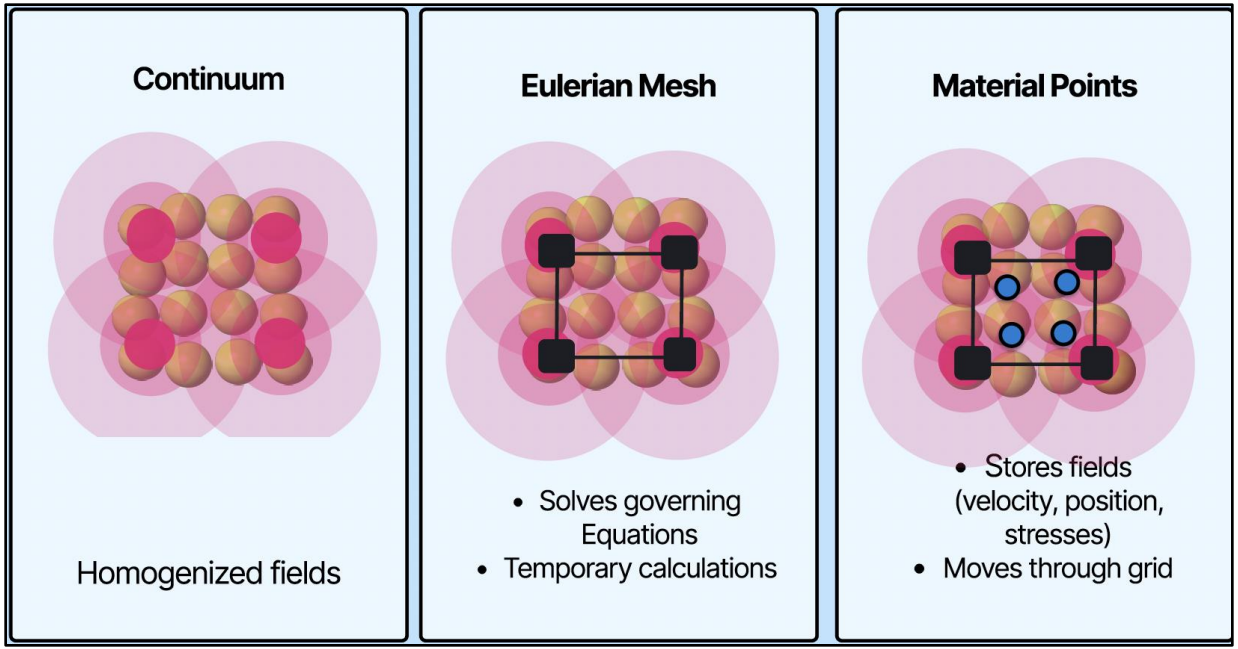
BOUNDARY VALUE PROBLEM

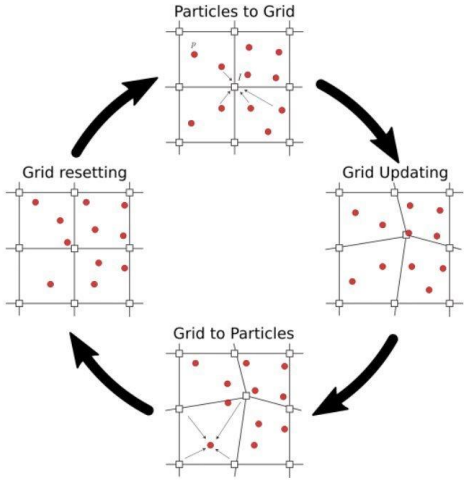
(GEOPHYSICAL GRANULAR FLOW)

MATERIAL POINT METHOD

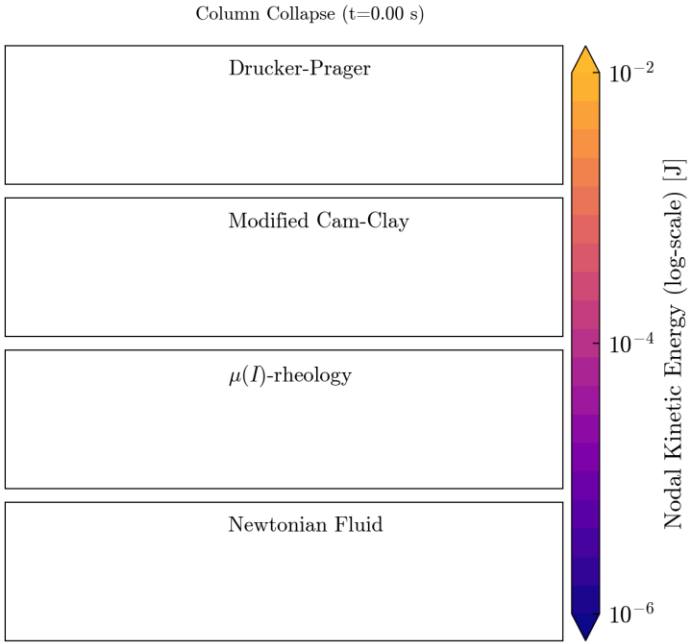


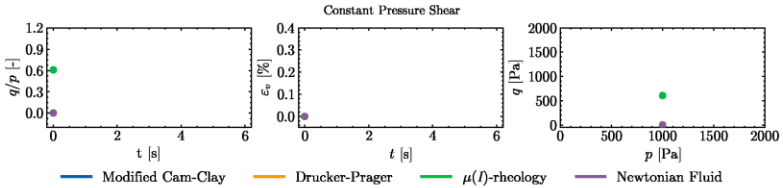
shape function





HydraxMPM





- Explicit Dynamic MPM
- AFLIP mapping
- Quadratic splines
- Gradient Aware simulations
- Batched simulations

https://github.com/GrainLearning/HydraxMPM

(SYNTHETIC) GEOPHYSICAL FLOW BENCHMARK

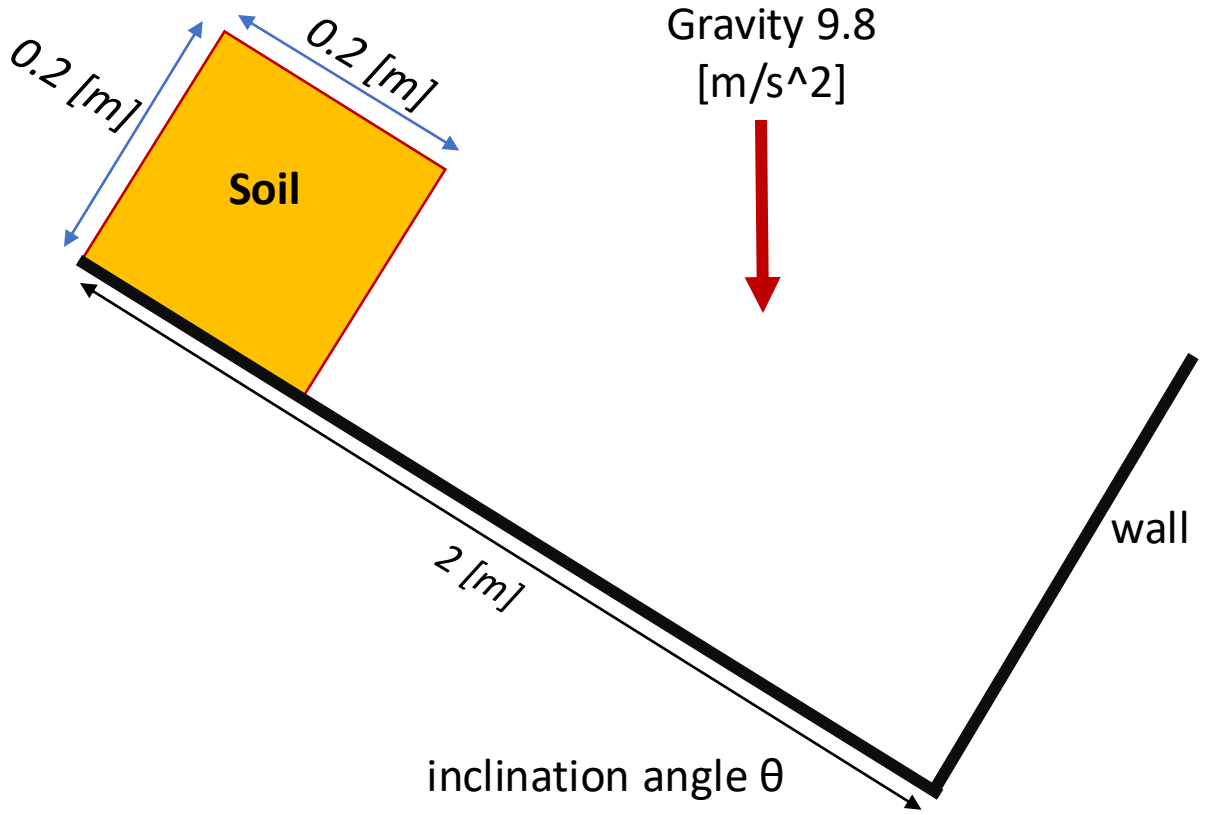
Simulation setup

Number of particles = *6400*
Total time = *3 [s]*
Snapshot time = *0.05 [s]*

Material parameters

Friction angle = *25 [deg.]*
Bulk density = *2650 [kg/m^3]*
Stiffness = *7e5 [Pa]*
Poisson's ratio =*0.3 [-]*

100 simulations ~ 1 hour (NVIDIA A10)



SPECTRAL SURFACE FLOOR

Signed Distance Field

Mean value

Number of harmonics = 16

$$k \in \left[\frac{2\pi}{\lambda_{\max}}, \frac{2\pi}{\lambda_{\min}}\right]$$

$$z(x) = z_{\text{base}} + \sum_{i=1}^{N} A_i \sin(k_i x + \phi_i)$$

Root Mean Square normalization

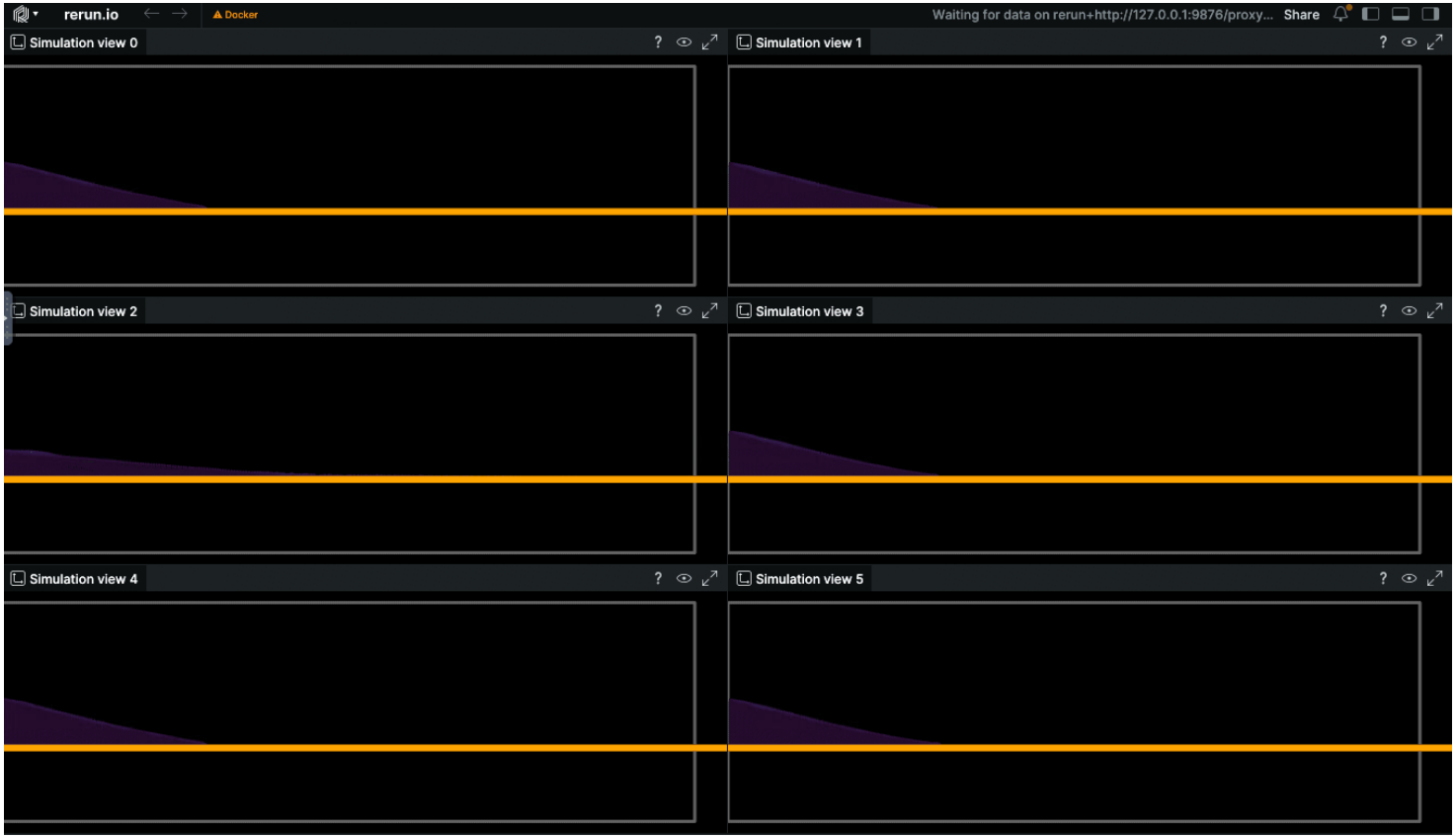
$$A_i^{\text{scaled}} = A_i \left(\frac{\text{Target RMS Height}}{A_{\text{RMS}}}\right) \qquad A_{\text{RMS}} = \sqrt{\sum_{i=1}^{N} \frac{A_i^2}{2}}$$

Random phase shift

Power-Law Scaling (Hurst Exponent)
to mimic natural geomorphic surfaces
H→1.0 (smooth)
H→1.0 (jagged)

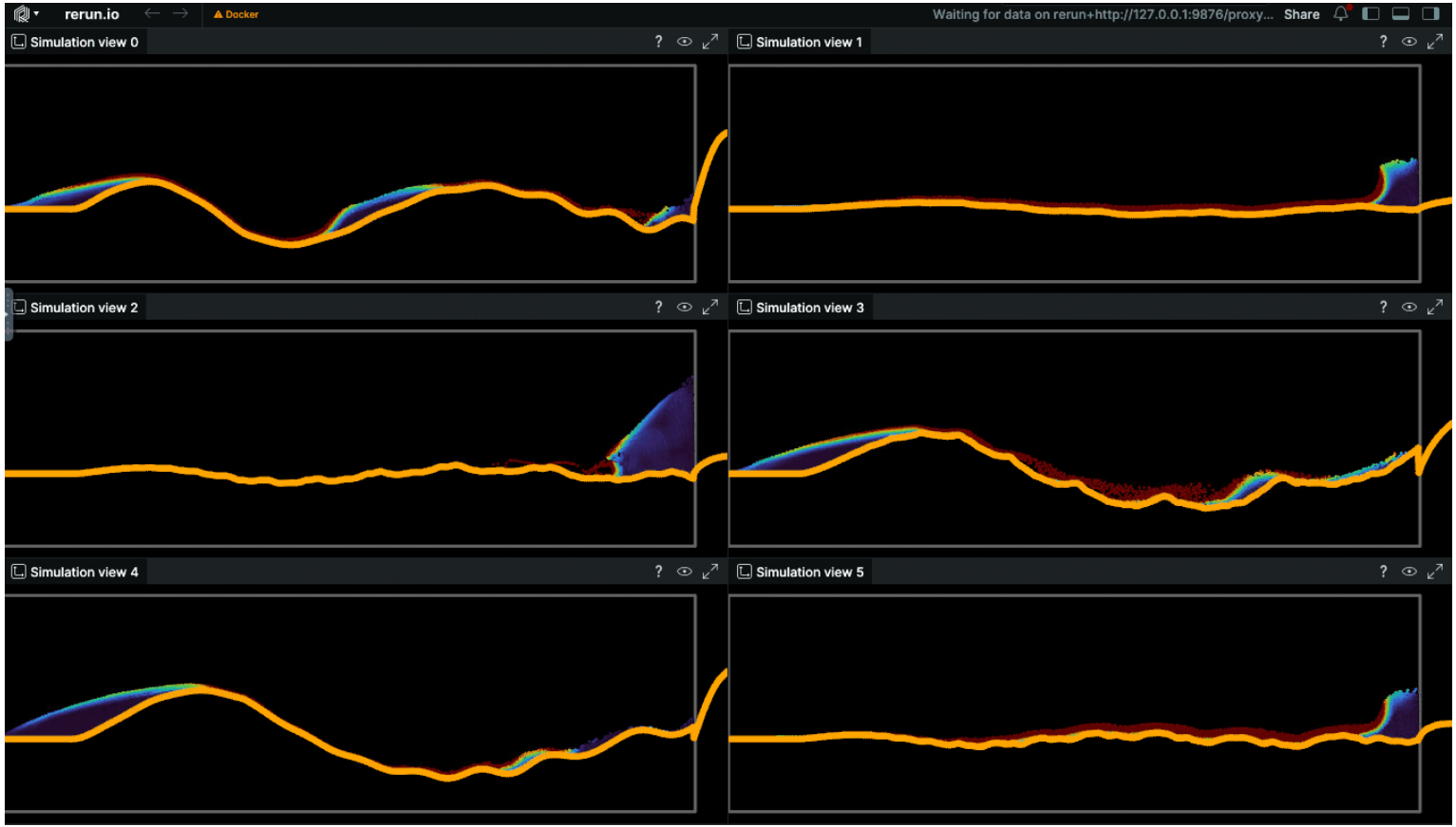
$$A(k) \propto k^{-(H+0.5)}$$

EXAMPLE VARYING FRICTION



Color map: velocity magnitude [0 to 0.5 m^2]

EXAMPLE VARYING SURFACE (INCLINED)



Color map: velocity magnitude [0 to 0.5 m^2]

BENCHMARK DATA

Data stored as zarr

- Lagrangian material point data
- Inclination angle θ [deg.]
- Spectral Surface Parameters
  - Geometry

$$H,\ \lambda_{\min},\ \lambda_{\max},\ A_{\mathrm{RMS}}$$

  - Friction

$$H,\ \lambda_{\min},\ \lambda_{\max},\ A_{\mathrm{RMS}}$$

Objective

- Learning the field evolution on moving material points

Example of material point data

shape [batch, snapshot, particles, dims..]

```
├── .F_stack (2, 39, 6400, 3, 3) float16
├── .L_stack (2, 39, 6400, 3, 3) float16
├── .force_stack (2, 39, 6400, 2) float16
├── .mass_stack (2, 39, 6400) float16
├── .position_stack (2, 39, 6400, 2) float16
├── .stress_stack (2, 39, 6400, 3, 3) float16
├── .velocity_stack (2, 39, 6400, 2) float16
├── .volume0_stack (2, 39, 6400) float16
└── .volume_stack (2, 39, 6400) float16
```

Note

- Possibility to train & generate data on the fly (via JAX)
- HydraxMPM utilities can be reused (e.g., rerun viewer)

CONCLUSION

Two benchmarks

Stage 0: Local element test benchmark

- Vary material parameters & predict stress-strain-volume response

Stage 1: Geophysical flow benchmark with MPM

- Vary boundary conditions & predict field evolution
- Floor (geometry and friction) sum of sine waves
- Vary angle of inclination

- **Data available on surfdrive** (https://surfdrive.surf.nl/s/m7EBKsKaNeiBqNT)
  password: hydraxmpm_ut
- **JAX-based MPM solver** (https://github.com/GrainLearning/HydraxMPM)
  `uv sync --extra full-cuda`

THANK YOU